Lab Summary: Blockchain Consensus Mechanisms

1 Learning Objectives

By the end of this lab sequence, students will be able to:

1. **Analyze Distributed Consensus:** Visualize how the "Longest Chain Rule" (Nakamoto Consensus) resolves competing histories (forks) and determines which blocks become canonical versus orphaned.
2. **Contrast Security Models:** Differentiate between the physical costs of Proof-of-Work (energy/hardware) and the internal economic penalties of Proof-of-Stake (slashing).
3. **Evaluate Centralization Risks:** Observe how economic factors—specifically operating costs and compound interest—create natural centralization pressures ("Rich Get Richer").
4. **Simulate Adversarial Strategies:** Actively deploy **Selfish Mining** (PoW) and **Block Withholding** (PoS) to understand how timing and strategy can be leveraged to exert disproportionate influence and reap additional rewards.

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2 Lab Module 1: Proof-of-Work (The Mining Simulator)

This module focuses on the mechanical propagation of blocks and the resolution of competing histories in a permissionless environment.

2.1 Activity A: Fork Resolution & Reorgs

- **Simulation:** Students run a local mining client, competing against simulated CPU miners to solve Hashcash puzzles.
- **Key Mechanics:**

  - **Competing Histories:** Students experience the uncertainty of distributed consensus firsthand. They may successfully mine a block, only to see it discarded moments later because a competing branch (fork) produced more work (a "Reorg").
  - **Visualizing Orphans:** The simulation concludes with an ASCII tree visualization, allowing students to clearly see the "branches" that were pruned and identify exactly where their local view diverged from the final canonical chain.

- **Objective Alignment:** Supports *Objective 1* by making the abstract concept of "probabilistic finality" concrete—students see their "confirmed" blocks turn into orphans.

2.2 Activity B: Strategic Withholding (Selfish Mining)

- **Simulation:** Students explore the "Selfish Mining" attack vector from two perspectives:
  1. **Adversarial Configuration:** Configuring a CPU bot to withhold valid blocks and "ambush" the network with a selfish mining strategy.
  2. **Interactive Strategy:** The student is prompted to manually decide when to publish their own mined blocks while controlling ~30% of the hashrate, attempting to bait honest miners into wasting work on a stale chain.

- **Insight:** Students learn that a miner with **less than 50% hash power** can still earn **more than their fair share** of rewards by strategically timing block releases to invalidate the work of honest competitors.
- **Objective Alignment:** Supports *Objective 4* by demonstrating that protocol compliance is not always the most profitable strategy (Nash Equilibrium).

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3 Lab Module 2: Proof-of-Stake (The Economic Simulator)

This module moves away from hashing to focus on key aspects of proof-of-stake systems.

3.1 Scenario A: The "Rich Get Richer" (Dilution Sim)

- **Activity:** A high-speed simulation (100k epochs) where validators have fixed operating costs (server fees).
- **Insight:** Students observe that while small validators generate profit, they cannot keep up with the compounding inflation of the "Whale" validators. The small players' percentage ownership of the network shrinks over time (dilution).
- **Objective Alignment:** Supports *Objective 3* by demonstrating the inherent centralization forces in PoS systems that lack redistribution mechanisms.

3.2 Scenario B: The Consensus War (Incentive Design)

- **Activity:** Students act as validators and encounter "Cartel" fork attempts. They face two distinct pressures:
    1. **The "Nothing at Stake" Dilemma:** The temptation to double-sign on both forks to guarantee rewards.
    2. **The Corruption Attempt:** Receiving bribes to support a malicious fork.
- **Key Insight:** This scenario highlights that corruption requires the attacker to *know* who the next validator is in advance to target them with a bribe.
- **Discussion Point:** This reinforces the critical importance of **Verifiable Random Functions (VRF)** and rapid validator rotation to prevent targeted collusion.
- **Objective Alignment:** Supports *Objective 2* by illustrating how Slashing conditions and Randomization work together to secure the chain.

3.3 Scenario C: Strategic Dominance (The Oligarch)

- **Activity:** Students observe a dominant stakeholder (70% stake) utilizing a **Block Withholding** strategy.
- **Key Mechanics:**
    - **The Ambush:** The Oligarch plays by the protocol rules (only proposing when selected) but uses their weight to secretly build a heavier chain, releasing it only to override honest proposals.
    - **Optimized Yield:** Rather than just censoring, the student observes how the Oligarch uses this timing advantage to orphan honest blocks, thereby capturing a larger share of the total network rewards than their stake alone would dictate.
- **Objective Alignment:** Supports *Objective 4* by proving that majority stake dominance grants the power to leverage **timing and strategy** to amplify influence and rewards beyond simple ownership percentage.